

Mongoose Aggregation Guide

Aggregation Pipeline: sequence of stages.

Key stages:

\$match - filter documents.

\$group - group documents using accumulators:

\$sum, \$avg, \$min, \$max, \$first, \$last, \$push, \$addToSet.

\$project - select, rename, compute fields.

\$sort - sort ascending(1)/descending(-1).

\$limit - limit results.

\$skip - pagination.

\$count - count documents.

\$lookup - join another collection.

\$unwind - deconstruct arrays.

\$addFields - add computed fields.

\$unset - remove fields.

\$replaceRoot - replace current document.

\$facet - multiple pipelines at once.

\$bucket / \$bucketAuto - group into ranges.

\$sample - random documents.

\$geoNear - geospatial search (must be first stage).

\$merge - write to existing collection.

\$out - write to new collection.

\$unionWith - union another collection.

Useful expressions:

\$concat, \$toUpper, \$toLower, \$ifNull, \$cond, \$switch,

\$multiply, \$divide, \$add, \$subtract, \$round,

\$size, \$filter, \$map, \$reduce.

Typical order:

\$match -> \$group -> \$sort -> \$project

Example:

```
Farm.aggregate([
  {$group: {_id: "$city", farmsCount: {$sum: 1}, averagePrice: {$avg: "$price"}}},
  {$sort: {averagePrice: -1}},
  {$project: {_id: 0, city: "$_id", farmsCount: 1, averagePrice: 1}}
])
```